

Certified Adversarial Robustness via Randomized Smoothing

Cohen, Rosenfeld, Kolter (2019)

Paper Presentation: Lanzeray Etienne, Ruelle Malo

The Problem:

- Classifiers achieve high accuracy on test sets
- **But:** Not robust to small adversarial perturbations
- Adversary finds δ where:
 - $\|\delta\|_2$ is small
 - $f(x + \delta) \neq f(x)$

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Defense Approaches:

- 1 **Empirical:** Adversarial training
 - × Often "broken" by stronger attacks
 - × No guarantees
- 2 **Certified:** Provable robustness
 - ✓ Verify constant prediction in neighborhood
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Adversarial Attacks & Robustness

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Key Challenge

Existing certified defenses don't scale to large networks (ImageNet classification)

What is Certification?

Goal

For classifier g , input x , and radius r , **verify** that:

$$g(x + \delta) = g(x) \quad \forall \|\delta\|_2 < r$$

This paper's contribution:

- ✓ First certified defense that **scales to ImageNet**
- ✓ Works with **any** base classifier
- ✓ Provides **tight** ℓ_2 guarantees

Randomized Smoothing: Core Idea

Definition

Given base classifier $f : \mathbb{R}^d \rightarrow \mathcal{Y}$ and noise level σ :

$$g(x) = \arg \max_{c \in \mathcal{Y}} P(f(x + \varepsilon) = c) \quad \text{where } \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Intuition:

- Add large Gaussian noise to input
- Return the class that appears most often under noise
- Large noise "drowns out" small adversarial perturbations

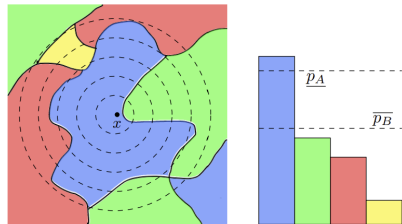


Figure: Evaluating the smoothed classifier at an input x .

Main Theoretical Result

Theorem (Tight Robustness Guarantee - Theorem 1)

If for class c_A and probabilities $\underline{p}_A, \bar{p}_B$:

$$P(f(x + \varepsilon) = c_A) \geq \underline{p}_A \geq \bar{p}_B \geq \max_{c \neq c_A} P(f(x + \varepsilon) = c)$$

Then $g(x + \delta) = c_A$ for all $\|\delta\|_2 < R$, where:

$$R = \frac{\sigma}{2} \left(\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\bar{p}_B) \right)$$

With Φ^{-1} the inverse of the standard Gaussian CDF.

Tightness of the Guarantee

Theorem (Theorem 2 - Tightness)

For any δ with $\|\delta\|_2 > R$, there exists a classifier f^ consistent with the probabilities such that $g(x + \delta) \neq c_A$.*

Interpretation:

- Gaussian smoothing **naturally induces** ℓ_2 robustness
- The certified set is *exactly* an ℓ_2 ball
- Cannot improve the bound without more assumptions!

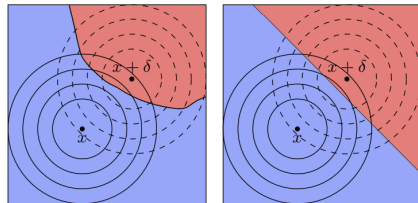


Figure: Worst-case bound

Practical Challenge: Monte Carlo Algorithms

Problem: Cannot exactly compute $P(f(x + \varepsilon) = c)$ when f is a neural network!

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- Sample n noisy predictions
- Count top two classes
- Use binomial test
- Return class or abstain

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Algorithm: CERTIFY

- Phase 1: Guess top class (n_0 samples)
- Phase 2: Estimate $\underline{p}_A$ (n samples)
- Compute $R = \sigma\Phi^{-1}(\underline{p}_A)$
- Return class + radius

Guarantee: With probability $\geq 1 - \alpha$, g robust within returned R

Experimental Results

ImageNet Results (ResNet-50):

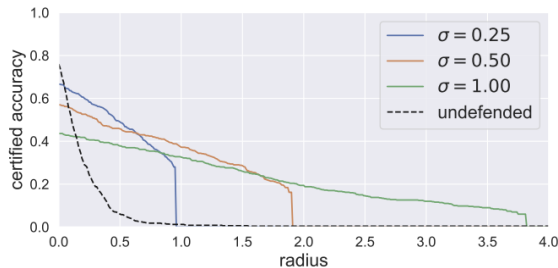


Figure: Approximate certified accuracy attained by randomized smoothing on ImageNet.

Key observations:

- First certified defense on full ImageNet!
- σ controls robustness/accuracy trade-off

Going Further: Other Noise Distributions

Natural Question

Gaussian noise $\Rightarrow \ell_2$ robustness. What about other ℓ_p norms?

Other Noise Distribution

Different noise distributions may induce robustness to different perturbation sets

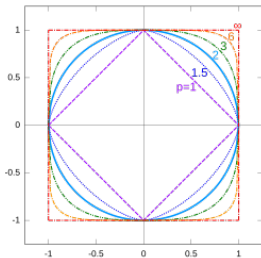


Figure: ℓ_p norms

Positive aspects:

- Clear theoretical analysis with complete proofs
- Code provided on Github
- Detailed appendices covering edge cases

Negative aspects:

- Intro with undefined terms